Syllabus

Course: BUS622 - DEVELOP TOOLS IN BUSINESS ANALYTICS
Goals: R AND PYTHON PROGRAMMING FOR MANAGERS

INSTRUCTOR INFORMATION

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COURSE INFORMATION

Meeting room: Bloomberg Research Center Lab
Time: 6:15PM - 9:00PM on Mondays
Office hours: by appointment only

COURSE DESCRIPTION

This course introduces students to various tools in business analytics development and reporting. Students are introduced to the basics of programming with R and Python to develop a working knowledge of how to acquire, manage, and treat data.

LEARNING OUTCOMES

- Understand and apply the main concepts used in data science and business analytics using functions and tools developed in Python
- Understand and use R functions in business data analysis
- Understand and apply basic coding and programing with tools such as R and Python

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COURSE MATERIALS

Texbook: no assigned textbook; readings, sample code, and assignments provided by the instructor via *Canvas*
Datasets: provided by the instructor

COMPUTING NEEDS

The Bloomberg Lab will be open during regular business hours and is equipped with all the necessary software for the course. **If you plan to attend any class online, it is your responsibility to have a computer with all required software installed.** If you need assistance installing R or Python, please contact the BW IT department.

GRADING

Grade Categories	**Teams / Individual**	**%**
Homework Assignments	Team	70
- *Peer Review Report*	Individual	0
- *Statement of AI Use*	Individual	0
- *Producing desired output*	Team	50
- *Class Presentation, if any*	Individual	20
Pop Quiz	Indivudal	10
Attendance + Class Participation	Individual	10
AI log + AI reflection statement	Individual	10

HOMEWORK ASSIGNMENTS

- All homework assignments must be completed **individually** and submitted **online via Canvas** by the *due date*. You may collaborate and sit together as a team to help each other with assignments, but the bulk of the work must be your own.
- Each homework assignment requires a statement of AI use.
- Each homework assignment requires a peer review report detailing the amount of help obtained or given to other team members.
- Homework will not be graded unless all individuals submit their statement of AI use and all team members submit the peer review report.
- Points will be deducted for late submissions or incomplete work. NO EXCEPTIONS.
- Instructions on which files to upload will be given during the time at which the homework is assigned. If you're unfamiliar with how to upload homework assignments on Canvas, please schedule an appointment with the BW IT department for assistance.

AI log + AI Reflection statement

USE OF AI FOR CODING

The *ethical* use of AI as a coding companion is encouraged. Its usage should align with the course's learning standards and should not serve as a substitute for the coding skills taught in class. Put simply, if you do not understand the code that AI recommends, DO NOT USE IT. There are explicit points for presenting your homework during class, and if you are unable to explain your own code, you will receive a ZERO for that assignment.

- Proper logging is mandatory and will help in writing the AI Reflection statement at the end of term.
  - Log every AI interaction during the assignment.
  - *What were you trying to accomplish?* (e.g., "Debug syntax error," "Understand how loops works," "Generate boilerplate code for a class")
  - *Which tool did you use?* (e.g., ChatGPT, Claude, Gemini, GitHub Copilot, other)
  - *Type of Assistance*
    - ☐ Explaining a concept
    - ☐ Debugging/fixing errors
    - ☑ Code generation (wrote new code)
    - ☐ Code review/optimization
    - ☐ Research/documentation lookup
    - ☐ Other: _____
  - *What did you learn from this interaction?* Did it help you understand the concept better, or just solve the immediate problem, or did it confuse you more?
  - *Approximate time on this AI interaction*: _____ minutes
  - AI work is highly detectable in homework. Using AI and ***not*** citing it will earn a ZERO for that homework. This detection will be compared to your Statement of AI Use to determine any grading penalty. This strictness reflects the university's policy on academic integrity: it is impossible to assign a grade for homework that does not reflect your own contribution but rather that of an AI, especially if you do not log it.
  - End of term, you are expected to turn in a detailed reflection on overall AI use patterns during the course. You may draw insights from your individual homework logs to write this report.

ATTENDANCE POLICY

Attendance is **mandatory** and **essential** for success in this course. You can either attend class in person or online via Zoom. If you are unable to attend live during class times, you *may* watch the class recordings via *Panopto* in *Canvas*, but you will forfeit attendance points for the live sessions you miss. You may regain those lost attendance points by emailing me at spaul@bw.edu a summary (5-page Word document) of your understanding of what was taught in class after watching the class recording. This summary email must be received before the start of next week's class, as shown by the email timestamp. Each class is tailored to help you successfully complete the homework assignments, which make up a substantial portion of your grade. Deadlines for homework will not be extended just because you were absent from class. In addition, class participation carries crucial grade points. You should make every effort to attend each class during class times unless you have a medically certified illness, a death in the family, or a University-sponsored activity. You are responsible for keeping up with all lecture material and announcements made in class.

COURSE SCHEDULE (tentative topics)

- *Python*
  - ☐ Introduction and basic console coding *Python*
  - ☐ Introduction to *Python* using *jupyter notebooks* and *py* scripts.
  - ☐ Handling data types: numeric, string, lists, dictionaries, dataframes
  - ☐ Loops: while, for, if
  - ☐ Importing and exporting datasets from varied sources (csv, json)
  - ☐ Basic calculations using *numpy* and *pandas*
    - ☐ Create
    - ☐ Replace
    - ☐ Update
    - ☐ Delete
    - ☐ Sort (Arrange)
    - ☐ Date treatment
  - ☐ Functions: validation and safe programming
  - ☐ performing a simple DCF using python code. (optional)
- *R*
  - ☐ Introduction to *dplyR*
  - ☐ Import and Save datasets
  - ☐ Simple *SQL* programming with *dplyR*
  - ☐ Creating *R* Scripts
  - ☐ Using *dplyR* features
    - ☐ Create
    - ☐ Subset
    - ☐ Mutate / Transmute
    - ☐ Filter
    - ☐ Sort (Arrange)

        - [ ] Grouping and Summarise
        - [ ] Top N
    - [ ] Treatment of missing data
    - [ ] Miscellaneous R topics
        - [ ] Functions
        - [ ] Loops
        - [ ] Merging
        - [ ] Date treatment

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